Experiment 4: Design Custom Visualizations and Interactive Dashboards in Power BI.

Aim:

To design custom visualizations and create an interactive dashboard in Power BI using filters, transformations, and visual enhancements to analyze data effectively.

Theory:

Data visualization represents complex data in graphical form, making it easier to understand patterns, trends, and relationships. **Power BI** is a business intelligence tool used to create interactive reports and dashboards.

A dashboard displays multiple visualizations such as bar charts, line charts, pie charts, and tables in a single view, helping users quickly analyze information. Power BI also provides features like filters, slicers, and cross-filtering for interactive data exploration.

Before creating visualizations, data is cleaned and transformed using **Power Query Editor** by removing unnecessary columns, correcting data types, and handling missing values.

A well-designed dashboard presents information clearly and helps users make better decisions.

Procedure:

1. Open Power BI Desktop.
2. Click Home → Get Data and import a dataset (Excel/CSV).
3. Click Transform Data to open Power Query Editor.

4. Clean and prepare the dataset by:
    - Removing unnecessary columns
    - Renaming columns
    - Changing data types if required
5. Click Close & Apply to load the data.
6. Create visualizations such as bar charts, line charts, pie charts, and tables using the Visualizations pane.
7. Add slicers and filters to allow interactive data selection.
8. Customize visuals by adding titles, colors, legends, and labels.
9. Arrange multiple visuals on the canvas to create an interactive dashboard.
10. Save the Power BI report.

Conclusion:

In this experiment, custom visualizations and an interactive dashboard were created using **Power BI**. The dataset was cleaned and transformed before generating charts and visuals. Filters and slicers were used to make the dashboard interactive..